



## LOCALLY SOURCED MICROGREENS

FINAL PROJECT REPORT

ENTR 390 001 WINTER 2024

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## Introduction

Fourth Soil Farm is a locally-grown, organic microgreen company serving the Ann Arbor area. Microgreens are “living foods” that are purposefully utilized in their most nutritious and delicious state. There are many health benefits to the superfoods, and there are four varieties that Fourth Soil Farm provides — broccoli, sunflower, pea, and crunchy mix. We are targeting health-conscious Ann Arbor residents who want to improve their diet and the taste of their meals. Valuing excellent customer service, we take pride in providing shipping less than 24 hours after our microgreens have been harvested. In the future we hope to expand our already growing customer base and increase the commercial clients that we have.

## Business Plan

### Marketing Segmentation

Our market segmentations are individual consumers, notably health-conscious people and chefs of restaurants sourcing food for their dishes. Health-conscious people are more prone to buy produce, especially specialty produce like microgreens. Chefs from restaurants are the primary buyer of microgreens because of the vegetable’s contribution to flavor and aesthetics within a dish. There are roughly 400 Ann Arbor restaurants that Fourth Soil Farm can serve and begin to start partnerships with the chefs as buyers (<https://www.annarbor.org/restaurants/>). There are about 120k Ann Arbor residents and it is predicted that over 18k are vegetarian or vegan based on the national statistics (<https://www.city-data.com/city/Ann-Arbor-Michigan.html>). Roughly 4% of America is vegetarian and 1% is vegan (<https://news.gallup.com/poll/510038/identify-vegetarian-vegan.aspx>). This was the closest

predictor for measuring the population of health-conscious individuals in the city but believe this is a low estimate in Ann Arbor because it is ranked as one of the top vegan-friendly cities

(<https://www.clickondetroit.com/all-about-ann-arbor/2020/12/13/peta-ann-arbor-is-eighth-most-vegan-friendly-small-city-in-us/>).

Figure I: Market Segmentation

| <b>Market Segment Name</b> | <b>Market Segment 1:<br/>Health Conscious People</b>                             | <b>Market Segment 2:<br/>Chefs / Restaurants</b>            |
|----------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Task</b>                | Provide a nutritious and palatable product                                       | Provide quality sprouts for their dishes                    |
| <b>Benefit</b>             | Local, healthy, and fresh                                                        | Local, healthy, and fresh                                   |
| <b>Urgency of Need</b>     | Low urgency of need                                                              | Medium urgency of need                                      |
| <b>Example End User</b>    | Young adults who are trying to eat fresh, vegan, or vegetarian meals             | Chefs who want sprouts as part of their specialty dishes    |
| <b>Lead Customers</b>      | People who are foodies and have been exposed to a healthy and more creative diet | Chefs that like their meals to be fresh and locally sourced |
| <b>Frequency of Buying</b> | One package a week                                                               | One tray a week (4 packages)                                |
| <b>Primary Buyers</b>      | Ann Arbor healthy eaters                                                         | Local restaurants                                           |



|                                    |                                                                                            |                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Concentration of Buyers</b>     | 18,318                                                                                     | > 400                                                                                      |
| <b>Competitives / Alternatives</b> | White Lotus Farms, Green Things Farm Collective, Argus Farm Stop, Ann Arbor Farmers Market | White Lotus Farms, Green Things Farm Collective, Argus Farm Stop, Ann Arbor Farmers Market |
| <b>Important Partners</b>          | Ann Arbor Farmers Market, community groups                                                 | Prominent chefs and local restaurants                                                      |

## Competitor Analysis

Local farms will be Fourth Soil Farm's main competitors, especially because microgreens are an ideal product to grow on a small scale in terms of profitability and low production costs <https://grocycle.com/microgreens-for-profit/>. Both White Lotus Farms and Ann Arbor Farmers Market have a reputable following and Green Things Farm Collective has a loyal following considering their CSA is completely sold out until the fall. White Lotus Farms is also the first and second websites to pop up when looking up "Ann Arbor Microgreens" but Fourth Soil Farm is third. Overall, White Lotus Farms is the greatest competitor due to one of their primary focus points being microgreens; they also are one of the primary sellers at the farmers market indicating a merging of competition. Green Things is not as worrisome because they do not produce microgreens directly though they produce similar products. This indicates that their customers still might be open to purchasing Fourth Soil's microgreens. What differentiates us from our competitors and makes this market hopeful is that we are still the only seller who

produces and sells solely microgreens. The lack of diversification will allow us to put our energy in gathering customers specifically interested in microgreens.

Figure II: Competitor Analysis

|                                 | <b>Physical Product</b>                     | <b>Average Price</b> | <b>About Us</b>                                                                                                                          | <b>Website Content (about us page)</b>                               | <b>Social Media (# of followers)</b>         |
|---------------------------------|---------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------|
| <b>White Lotus Farms</b>        | Organic, multiple variations of microgreens | \$5                  | Here we grow immune boosting Microgreens, make handcraft artisan cheese and bake breads, and pastries with 100% organic, regional flour. | Organic farm, bakery, and skincare line                              | Instagram: 15.8k followers, 1-2 daily posts  |
| <b>Ann Arbor Farmers Market</b> | Microgreens from an assortment of farmers   | \$5-\$10             | “Providing the Ann Arbor Area with Top Quality Produce, Farm Products, Prepared Food, & Artisan Goods Since 1919!”                       | Produce, prepared food and drink, meat and dairy as well as artisans | Instagram: 16.1k followers, 2-5 posts a week |
| <b>Green</b>                    | Assorted                                    | \$80 or              | “Tending to land and                                                                                                                     | Produce and                                                          | Instagram:                                   |

|                                       |                                    |                              |                       |         |                                        |
|---------------------------------------|------------------------------------|------------------------------|-----------------------|---------|----------------------------------------|
| <b>Things<br/>Farm<br/>Collective</b> | produce and<br>greens CSA<br>style | more for<br>winter<br>greens | feeding a community.” | flowers | 5,423<br>followers, 1-2<br>posts daily |
|---------------------------------------|------------------------------------|------------------------------|-----------------------|---------|----------------------------------------|

### Financial Forecast

The forecast shows projections for Fourth Soil Farms over the course of 5 years. The business model is focused on becoming the prominent seller within the local community; assuming customer loyalty over one time purchases. From the growth displayed, it is predicted that restaurants will become a more prominent customer base within the next 5 years.

Figure III: Summary P&L

|                                                      | <b>Year 1</b> | <b>Year 2</b> | <b>Year 3</b> | <b>Year 4</b> | <b>Year 5</b> |
|------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|
| <b>Units Sold<br/>(per package<br/>not per tray)</b> | 2,787         | 9,984         | 21,840        | 24,933        | 72,800        |
| Units Sold to<br>Customers                           | 1,667         | 6,240         | 10,608        | 11,333        | 26,000        |
| Units Sold to<br>Restaurants                         | 1,120         | 3,744         | 11,232        | 13,600        | 46,800        |
| <b>Revenue</b>                                       | \$15,600      | \$56,160      | \$106,080     | \$136,000     | \$390,000     |
| COGS                                                 | \$7,942.95    | \$28,454.40   | \$62,244      | \$71,060.00   | \$207,480     |

|                             |            |             |          |             |           |
|-----------------------------|------------|-------------|----------|-------------|-----------|
| <b>Gross Profits</b>        | \$7,657.05 | \$27,705.60 | \$43,836 | \$64,940.00 | \$182,520 |
| General &<br>Administrative | \$500      | \$1,000     | \$2,000  | \$2,000     | \$6,000   |
| Sales &<br>Marketing        | \$250      | \$1,000     | \$12,000 | \$12,000    | \$24,000  |
| <b>Total Costs</b>          | \$750      | \$2,000     | \$14,000 | \$14,000    | \$30,000  |
| EBITDA                      | \$6,907.05 | \$25,705.60 | \$29,836 | \$50,940    | \$152,520 |

### Financial Details

Figure IV: Product Pricing and Sales Drivers

|                  | <b>Retail Price</b> | <b>Offered at</b> | <b># Units Sold</b> |
|------------------|---------------------|-------------------|---------------------|
| <b>Product 1</b> | \$6                 | Year 1            | 440                 |
| <b>Product 2</b> | \$6                 | Year 1            | 500                 |
| <b>Product 3</b> | \$6                 | Year 1            | 787                 |
| <b>Product 4</b> | \$6                 | Year 1            | 1060                |

Figure V: Discount Rate Per Order Size

|                   |     |                      |     |
|-------------------|-----|----------------------|-----|
| <b># of Units</b> | < 4 | <b>Selling Price</b> | \$6 |
|-------------------|-----|----------------------|-----|

|                   |               |                      |            |
|-------------------|---------------|----------------------|------------|
| <b># of Units</b> | <b>&gt; 4</b> | <b>Selling Price</b> | <b>\$5</b> |
|-------------------|---------------|----------------------|------------|

Figure VI: Growth Rate Based on Consumer Type

|                    | <b>Year 2</b> | <b>Year 3</b> | <b>Year 4</b> | <b>Year 5</b> |
|--------------------|---------------|---------------|---------------|---------------|
| <b>Consumer</b>    | 274%          | 70%           | 7%            | 129%          |
| <b>Restaurants</b> | 234%          | 143%          | 50%           | 244%          |

Figure VII: Expected Units Sold Based on Customer Type (Year 1)

| <b>Customer Type</b> | <b>Purchase Size</b>    | <b># of Customers</b> | <b>Avg. Order Size in Individual Units</b> | <b>Avg. Discount</b> | <b>Price per unit</b> | <b>Revenue (1 of Each Type)</b> |
|----------------------|-------------------------|-----------------------|--------------------------------------------|----------------------|-----------------------|---------------------------------|
| <b>Consumer</b>      | Buying Individual Units | 34.73                 | 4 / month                                  | \$0                  | \$6                   | \$288                           |
| <b>Restaurants</b>   | Buying Bulk (trays)     | 5.83                  | 16 / month                                 | \$1                  | \$5                   | \$960                           |

Figure VIII: Revenue Growth per Customer Type

|                      | <b>Year 1</b> | <b>Year 2</b> | <b>Year 3</b> | <b>Year 4</b> | <b>Year 5</b> |
|----------------------|---------------|---------------|---------------|---------------|---------------|
| <b>Consumer</b>      | \$10,000      | \$37,440      | \$63,648      | \$68,000      | \$156,000     |
| <b>Restaurants</b>   | \$5,600       | \$18,720      | \$45,432      | \$68,000      | \$234,000     |
| <b>Total Revenue</b> | \$15,600      | \$56,160      | \$106,080     | \$136,000     | \$390,000     |

Figure IX: Percentage of Revenue Based on Customer Type

|                    | <b>Year 1</b> | <b>Year 2</b> | <b>Year 3</b> | <b>Year 4</b> | <b>Year 5</b> |
|--------------------|---------------|---------------|---------------|---------------|---------------|
| <b>Consumer</b>    | 64%           | 66%           | 60%           | 50%           | 40%           |
| <b>Restaurants</b> | 36%           | 33%           | 40%           | 50%           | 60%           |

Figure X: Cost of Goods Sold (COGS)

| <b>Microgreens</b>                                      | <b>Product 1</b> | <b>Product 2</b> | <b>Product 3</b> | <b>Product 4</b> | <b>Average</b> |
|---------------------------------------------------------|------------------|------------------|------------------|------------------|----------------|
| <b>Cost</b>                                             | \$0.75           | \$0.75           | \$0.75           | \$0.75           | \$0.75         |
| <b>Packaging</b>                                        | \$0.65           | \$0.65           | \$0.65           | \$0.65           | \$0.65         |
| <b>Shipping to customer<br/>(\$3 cost split between</b> | \$0.75           | \$0.75           | \$0.75           | \$0.75           | \$0.75         |

|                               |        |        |        |                     |               |
|-------------------------------|--------|--------|--------|---------------------|---------------|
| <b>4 individual packages)</b> |        |        |        |                     |               |
| <b>Labor</b>                  | \$0.70 | \$0.70 | \$0.70 | \$0.70              | \$0.70        |
|                               |        |        |        | <b>Average cost</b> | <b>\$2.85</b> |

The costs of small-scale microgreen farms are very low, hence the attractiveness of getting into the business. Additionally, with delivery being “direct to community” as described by Fourth Soil owners, the shipping costs are quite low because they do it all themselves. With the current growth projections, it can be expected that Fourth Soil’s microgreens will reach further than solely Ann Arbor in the coming years but ideally, the \$.75 per unit delivery cost can remain the same.

### LTV/CAC Analysis

- Descriptions of following charts

Figure XI: LTV/CAC Analysis

|                          |        |     |             |            |    |
|--------------------------|--------|-----|-------------|------------|----|
| <b>CAC Calculations:</b> |        |     |             |            |    |
| CAC                      | MCC    | CA  |             |            |    |
| 75.98                    | 60,750 | 800 |             |            |    |
| <b>LTV Calculation:</b>  |        |     |             |            |    |
|                          | LTV    | AOV | FR (1 year) | CL (years) | GM |

|                |          |        |     |     |        |
|----------------|----------|--------|-----|-----|--------|
| Consumer       | x        | \$6    | 48  | 10  | 52.50% |
| Restaurants    | x        | \$5    | 192 | 5   | 41.35% |
| Total          | 2,305.70 | \$5.50 | 120 | 7.5 | 46.60% |
| <b>LTV/CAC</b> |          |        |     |     |        |
| <b>Ratio:</b>  |          |        |     |     |        |
|                | LTV/CAC  |        |     |     |        |
|                | Ratio    |        |     |     |        |
|                | 30.3     |        |     |     |        |

*CAC: Customer Acquisition Cost, MCC: Marketing Campaign Cost, CA: Total Customers*

*Acquired*

*LTV: Lifetime Value, AOV: Average Order Value, FR: Frequency Rate, CL: Customer Lifespan*

## Product

### Product Description

Fourth Soil Farm offers 4 varieties of locally-grown microgreens for purchase.

Microgreens are the earliest stages of a vegetable's development. Up to more than 40x more nutrient-dense than fully grown plants, microgreens are high in calcium, magnesium, phosphorus, and potassium. Eating microgreens as a consistent part of one's diet can boost one's immune system, lower inflammation, and improve gut health.

We supply our microgreens from an organic farm in Ypsilanti, Michigan. Personally grown by Karl and Jeanne Thuemmel, the owners value a local, "direct to community" approach focusing on increasing microgreen consumption in the Ann Arbor area. We currently offer 4



individual microgreen varieties: broccoli, sunflower, pea, and crunchy mix. These can be purchased individually or in a weekly variety pack on our website. Because of the local focus of the company, sourcing the microgreens from a nearby farm allows for a customer-focused approach. Sourcing of the microgreens is all done within the scope of Fourth Soil Farm. The co-owners grow the microgreens themselves and therefore are easier to maintain and monitor. All of the supplies necessary to grow the microgreens are in addition to direct to consumer, we also are focusing on expanding to restaurants in the Ann Arbor area who want to use microgreens in their dishes.

After purchasing, fresh microgreens are delivered to customers' doorsteps within 24 hours of harvesting. The products are then driven and delivered to customers' homes by company employees. Being that the focus of the company is customers in the Ann Arbor area, we are able to provide free, express shipping both direct to customers and businesses.

Figure XII: Product Descriptions

| Product Number | Product Image                                                                       | Product Description  | Product Quantity | Product Cost |
|----------------|-------------------------------------------------------------------------------------|----------------------|------------------|--------------|
| 1              |  | Broccoli Microgreens | 2 oz             | \$6.00       |
| 2              |  | Sunflower            | 3.5 oz           | \$6.00       |

|   |                                                                                    |                           |        |         |
|---|------------------------------------------------------------------------------------|---------------------------|--------|---------|
| 3 |   | Pea Microgreens           | 3.5 oz | \$6.00  |
| 4 |   | Crunchy Mix               | 2.5 oz | \$6.00  |
| 5 |  | Weekly micro-variety pack | Varies | \$20.00 |

## E-Commerce Services

We have opted to use Wix for our choice of e-commerce. Utilizing this website will enable us to further engage with our established customers while also assisting in our future business development. We valued making a website that was easy to navigate, informative, and would lead to an increase in sales.

During the development of our website, we focused on having website pages that would engage our customers and direct them towards purchasing our microgreens. In addition to a home page, we have “Our Vision” to provide customer testimonials and shed light into our founders’ mission. The Fourth Soil Farm website naturally has a “Shop” page for customers to learn more about our product and complete their purchases. Lastly, we have a “Blog” and

“Contact Us” page to help engage with our current and potential customers both on our website and social media pages.

### **Website Pages**

**Home:** The landing page with product photos, an “Order Now” button directing customers to our “Shop” page, and a Welcome area where customers are directed to “Our Vision.” In addition to this, when customers open the home page, there is a pop-up in which they can sign up to be part of our informational emails.

**Our Vision:** Includes information about our founders and the overarching goals of Fourth Soil Farms. There are multiple customer testimonials on this website as well as information on the harvesting and shipping processes.

**Shop:** The shop page is naturally where customers can browse the various products for sale and make their selections. We not only included images of the microgreens but provided descriptive information on the health benefits of each. The prices and quantities of the products are clearly seen and easily navigable.

**Blog:** Our “Blog” page is where we hope to engage current customers with both our product and each other. Employees and customers alike can make posts about current recipes they love, questions they may have, or any other general health and wellness tips. Emphasizing this aspect of the website will improve our Search Engine Optimization as we can target customers using keywords found in our blog.

**Contact us:** This page is not only where customers can see the location of the farm itself, but they can easily send us direct messages with a simple email address, subject, and message. This gives people the opportunity to ask questions and provide feedback while also providing legitimacy.

## **Integrated Applications**

In order to increase overall customer engagement, we have created links at the top of the page to direct customers to our specific social media platforms. These are clearly shown on the top left of the webpage and are easily identifiable by their icons. This allows customers to further explore all that Fourth Soil Farm has to offer and provides a place for customers to interact with each other and us.

## **Marketing**

### **Social Media**

We decided to select platforms tailored specifically to the target audience of Fourth Soil Farm: Facebook and Instagram. Facebook is mainly utilized for its Marketplace, Groups, Live events, and targeted Ads, reaching more family-oriented consumers, predominantly ranging from ages 24-64. It's an ideal space for creating a “Fourth Soil Farm Community” group to encourage discussions on recipes and education about microgreens, as well as serve as a customer support forum. Instagram mainly captivates users who are looking for more visuals and storytelling through Stories, Highlights, and Reels to share more daily activities and customer testimonials, primarily young adults.

One post was created on Instagram and was for the weekly “Microgreen Monday” posts, a Fourth Soil Farm original challenge, that encourages customers to learn and engage with new and familiar microgreens. The other post was for Facebook, and it was of a cell phone with the Caller Id “Happier You” calling the customer, with the caption discussing why it is important to choose microgreens over others.

The two main competitors in the local microgreens social media scene that we've analyzed were White Lotus and the Ann Arbor Farmers Market. Both competitors are successful in their social media presence with inviting, clean visuals, and engaging content that emphasizes a farm-to-table approach. Taking inspiration from their strategy, we aim to sprinkle in our unique touch, leveraging our farm's story and the in-depth educational angle we provide about the benefits of microgreens. What sets Fourth Soil Farm apart is our commitment to sustainable, organic farming practices and personalized, direct community engagement. With each post, we aim to not just sell but to foster a lifestyle.

Figure XIII: Instagram Post



Figure XIV: Facebook Post

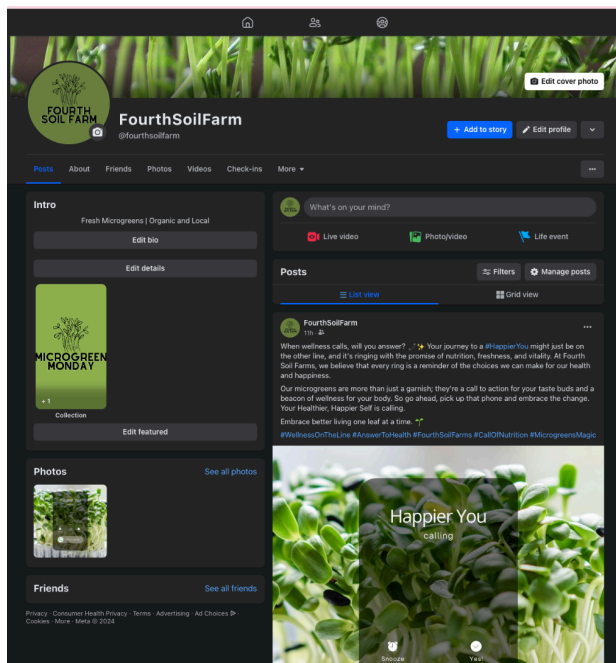
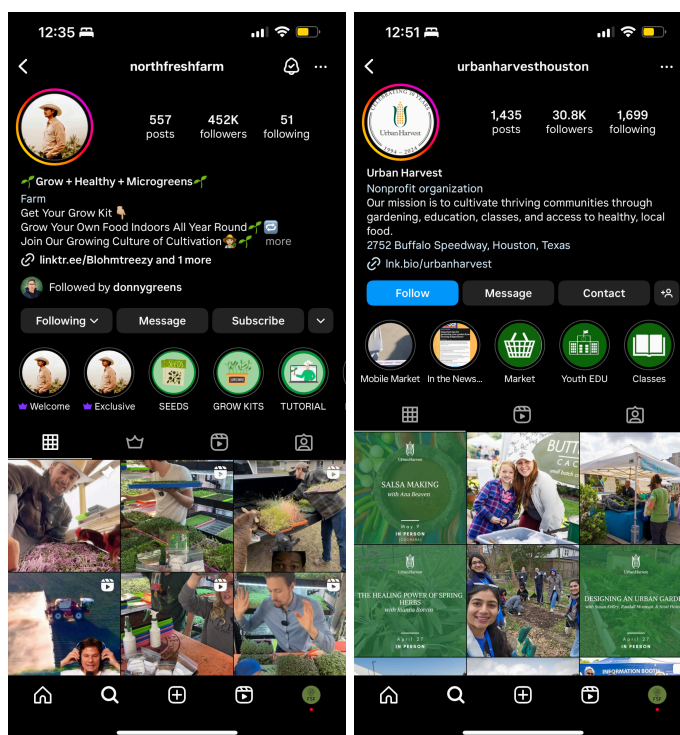


Figure XV: Our competitors' Instagram Pages



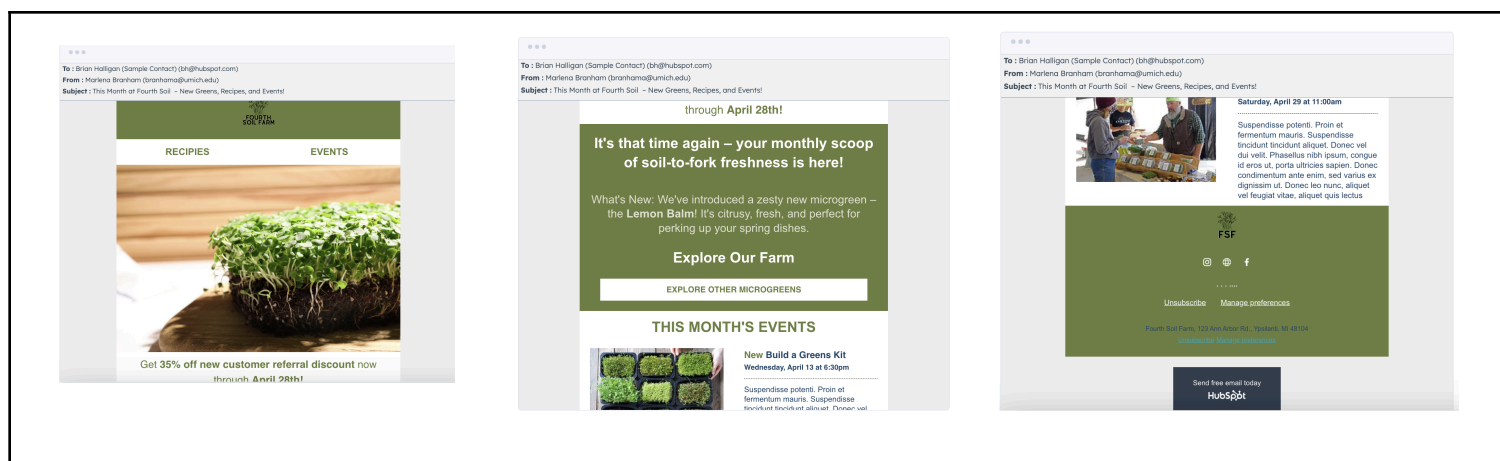
## Email Marketing

Utilizing email marketing is crucial in engaging current and potential customers. In order to achieve our goal of specifically increasing revenue from the business-to-business model, email marketing using HubSpot will help us engage with potential commercial customers.

We have a personalized welcome email series for our new subscribers, designed to acquaint them with the farm, highlight the benefits of microgreens, and provide exclusive introductory offers. In addition to the welcome series, our subscribers can look forward to regular newsletters featuring farm updates, new product launches, special recipe inspirations, and subscriber-exclusive promotions. Our emails are optimized for both desktop and mobile viewing, ensuring a seamless experience across all devices. Placed strategically within the email is a prominent Call To Action Button, inviting users to explore our microgreen collection directly on our website.

Towards the bottom of each email, subscribers will find a variety of local events and farm news, with links for registration through the email. As an added incentive, a special discount offer is displayed above the Call To Action Button, providing users with savings before they even begin their shopping journey. We respect our customers' preferences by offering them the option to unsubscribe from our marketing communications and provide quick access to our website and social media platforms at the conclusion of every email to keep our subscribers connected with Fourth Soil Farm on their preferred platforms.

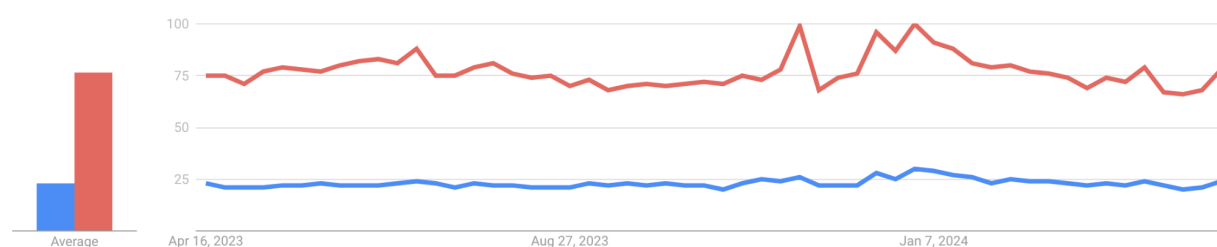
Figure XVI: Email Marketing



## Search Engine Optimization (SEO) and Keywords

Utilizing Wix's tools have made beginning search engine optimization approachable and likely to succeed. Given the local nature of the company, we focused most of our Google Trends research in the Michigan area. For example, “microgreens,” naturally one of our key words, has relatively high search popularity compared to other states. As shown below in figure \_\_, although we initially placed “vegetarian” as a keyword, after completing research on what users are more commonly searching for, we opted to use “vegan” because of its steadily high popularity. In addition to these words, we also used “Ann Arbor” and “organic” to gain attention from health-conscious consumers near Fourth Soil Farm.

Figure XVII: Google Search Trends Vegan (Red) vs. Vegetarian (Blue)





## **Conclusion**

Our business has an up-and-running inventory, an increasing customer base as well as an increasing product inventory. With our Facebook social media platform, our Instagram platform that will continue to become more active, and an email marketing campaign, we will be able to successfully make more purchases. Utilizing this new website will allow Fourth Soil Farm to further engage with current customers while allowing future buyers to learn more about the benefits of locally-grown microgreens. At the end of 5 years, we anticipate the majority of our customers coming from Ann Arbor area restaurants. An increase in social media activity and targeted emails will allow us to reach the goal of expanding to commercial clients.